$$C_{P,V} = \frac{q}{\Delta T}$$

$\Delta E = q + w$ (Zumdahl uses E, most others use U)

$|w = P\Delta V|$

$\Delta H = \Delta E + \Delta(PV)$ $q_p = \Delta H$ $q_V = \Delta E$

$$\Delta H^o_{reaction} = \sum \Delta H^o_f(products) - \sum \Delta H^o_f(Reactants)$$

Table 1: *Average Bond Energies (kJ/mol)*

Single Bonds						Multiple Bonds	
H–H	432	N–H	391	I–I	149	C=C	614
H–F	565	N–N	160	I–Cl	208	C≡C	839
H–Cl	427	N–F	272	I–Br	175	O=O	495
H–Br	363	N–Cl	200			C=O*	745
H–I	295	N–Br	243	S–H	347	C≡O	1072
		N–O	201	S–F	327	N=O	607
C–H	413	O–H	467	S–Cl	253	N=N	418
C–C	347	O–O	146	S–Br	218	N≡N	941
C–N	305	O–F	190	S–S	266	C≡N	891
C–O	358	O–Cl	203	P-H	339	C=N	615
C–F	485	O–I	234	Si–Si	340		
C–Cl	339			Si–H	393		

	1 IA												13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1	1 H 1.008	2 IIA																2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.179
3	11 Na 22.989	12 Mg 24.305	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8	9 VIIIB	10	11 IB	12 IIB	13 Al 26.981	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.955	22 Ti 47.867	23 V 50.941	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.921	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.905	46 Pd 106.42	47 Ag 107.868	48 Cd 112.411	49 In 115.818	50 Sn 118.710	51 Sb 121.760	52 Te 127.60	53 I 126.904	54 Xe 131.29
6	55 Cs 132.905	56 Ba 137.327	57 La* 138.905	72 Hf 178.49	73 Ta 180.948	74 W 183.84	75 Re 186.207	76 Os 190.23	77 Ir 192.217	78 Pt 195.078	79 Au 196.967	80 Hg 200.59	81 Tl 204.383	82 Pb 207.2	83 Bi 208.980	84 Po (209)	85 At (210)	86 Rn (222)
7	87 Fr (223)	88 Ra (226)	89 Ac+ (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (264)	108 Hs (265)	109 Mt (268)	110 Unn (269)	111 Uuu (272)	112 Uub (277)						
8				*	58 Ce 140.112	59 Pr 140.908	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.964	64 Gd 157.25	65 Tb 158.925	66 Dy 162.50	67 Ho 164.930	68 Er 167.26	69 Tm 168.934	70 Yb 173.04	71 Lu 174.967
9				+	90 Th 232.038	91 Pa 231.036	92 U 238.039	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

$R = 8{,}3145\ J\ K^{-1}mol^{-1} = 0{,}08206\ dm^3\ atm\ mol^{-1}\ K^{-1}$

$F = 96485\ C\ mol^{-1}$

$0\ ^oC = 273{,}15\ K$

$1\ V = 1\ J\ C^{-1} = J\ A^{-1}\ s^{-1}$

$1\ atm = 101.\ 325\ kPa$

$= 760\ torr$

$1\ bar = 10^5\ Pa$

$1\ J = Pa\ m^3$

Compound	Formula	$H_f^{ø}$ kJ mol^{-1}
Carbon monoxide	$CO_{(g)}$	-110.5
Carbon dioxide	$CO_{2(g)}$	-393.5
Water	$H_2O_{(g)}$	-242
	$H_2O_{(l)}$	-286
Alkanes		
Methane	$CH_{4(g)}$	-74.8
Ethane	$CH_3CH_{3(g)}$	-84.7
Propane	$CH_3CH_2CH_{3(g)}$	-104.5
Butane	$CH_3(CH_2)_2CH_{3(g)}$	-126.5
Pentane	$CH_3(CH_2)_3CH_{3(l)}$	-173.2
Hexane	$CH_3(CH_2)_4CH_{3(l)}$	-198.6
Heptane	$CH_3(CH_2)_5CH_{3(l)}$	-224.0
Octane	$CH_3(CH_2)_6CH_{3(l)}$	-250.0
Nonane	$CH_3(CH_2)_7CH_{3(l)}$	-274.0
Decane	$CH_3(CH_2)_8CH_{3(l)}$	-300.9
Alcohols		
Methanol	$CH_3OH_{(l)}$	-239.1
Ethanol	$CH_3CH_2OH(l)$	-277.1
Propan-1-ol	$CH_3CH_2CH_2OH_{(l)}$	-302.7
Propan-2-ol	$CH_3CHOHCH_{3(l)}$	-317.9
Butan-1-ol	$CH_3(CH_2)_2CH_2OH_{(l)}$	-327.4
Pentan-1-ol	$CH_3(CH_2)_3CH_2OH_{(l)}$	-353.6
Hexan-1-ol	$CH_3(CH_2)_4CH_2OH_{(l)}$	-377.8